

# Hitesh Kumar Gupta

hiteshagarwal172@gmail.com • +91 7079051061 • github.com/ihiteshagrawal • linkedin.com/in/ihiteshagrawal

## EDUCATION

### K. K. College of Engineering & Management, Dhanbad

Expected May 2027

B.Tech in Computer Science & Engineering | **CGPA: 8.5/10 (First Rank)**

*Relevant Coursework: Data Structures & Algorithms, AI & Machine Learning, Information Retrieval, Data Mining, Operating Systems, Computer Networks, Databases, Discrete Mathematics*

## EXPERIENCE

### Indian Institute of Technology Kharagpur

May 2026 – Present

*R&D Intern | Under Dr. Koustav Rudra*

- Designed and implemented an end-to-end NLP pipeline: transcript cleaning, semantic chunking, QA generation, multi-gate quality filtering, near-duplicate deduplication, and leakage-safe video-level train/dev/test splitting.
- Generated QA pairs with Qwen2.5-14B-Instruct served via vLLM on an H100, ~100,000 raw pairs across 32,000+ semantic chunks from 1,100+ crypto YouTube videos.
- Coordinating the FIRE 2026 shared task track on cryptocurrency information retrieval, organized by IIT Kharagpur.

### Indian Institute of Technology (IIT) ISM Dhanbad

Feb 2026 – Apr 2026

*R&D Project Intern | Under Dr. Sachin Tripathi*

- Implemented a blockchain-assisted mutual authentication and key agreement scheme for fog-based healthcare infrastructure using Hyperledger Fabric, based on the BAoG protocol architecture.
- Configured a three-layer consortium network with cloud servers, fog nodes, and smart meter simulation; evaluated chaincode transaction performance and cryptographic overhead using ECC scalar multiplication.

### Birsa Institute of Technology, Sindri

June 2025 – July 2025

*Deep Learning Intern*

- Built and deployed production-ready deep learning pipelines for image processing, translating academic research into deployable software artifacts.
- Led a team of 4 engineers on model training pipelines and cloud deployment; achieved 100% deployment success rate across all AI systems.

### VaporFi Labs (Remote)

Sep 2022 – April 2026

*Software Engineer (Part-Time)*

- Architected a DEX aggregator across EVM chains; reverse-engineered closed-source routing logic to reconstruct optimal swap path algorithms minimising slippage and gas costs.
- Built event-driven blockchain data pipelines ingesting high-volume on-chain streams into PostgreSQL; optimised SQL indexes to sustain sub-200ms query response times under production load.
- Led end-to-end development of a high-volume NFT marketplace (65k+ assets) — ERC721 smart contracts, Merkle Tree allowlists, IPFS metadata storage, and React/Next.js frontend; improved Core Web Vitals from 63 to 95+ via ISR/SSG refactor.
- Developed REST and GraphQL APIs with Zod validation and TanStack Query caching, reducing redundant network calls by ~40%; set up Docker-based CI/CD pipelines for self-hosted infrastructure.

## PUBLICATIONS & RESEARCH

Gupta, H. K. (2025). **When Better Eyes Lead to Blindness: A Diagnostic Study of the Information Bottleneck in CNN-LSTM Image Captioning Models**. International Journal of Computer Applications. DOI: 10.5120/ijca2025925560

*Cited 3 times across IEEE-published research, IOP Science, and NCIT Journal.*

## TECHNICAL SKILLS

**NLP & ML:** LLM Evaluation, Fine-tuning, Transformers, PyTorch, TensorFlow, Agentic Frameworks, CNN/LSTM

**Languages:** Python, TypeScript, JavaScript (ES6+), SQL, Solidity

**Backend & APIs:** Node.js, FastAPI, Express.js, GraphQL, REST APIs, Event-Driven Architecture, WebSockets

**Databases:** PostgreSQL, Redis

**Cloud & Infra:** Docker, Kubernetes, CI/CD, Linux, Nginx

**Blockchain:** Solidity, Hyperledger Fabric, EVM

## **ACHIEVEMENTS**

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- **arXiv Endorsement (cs.CV, cs.LG):** Endorsed by established computer scientists for preprint publication — obtained independently without university affiliation.
- **First Rank — K. K. College of Engineering & Management:** Ranked first in the Computer Science & Engineering programme.
- **Hindustan Talent Honour 2023:** Recognised for outstanding academic excellence.